



Service Data

SD-29-50024

Bendix® FLC-25™ Camera

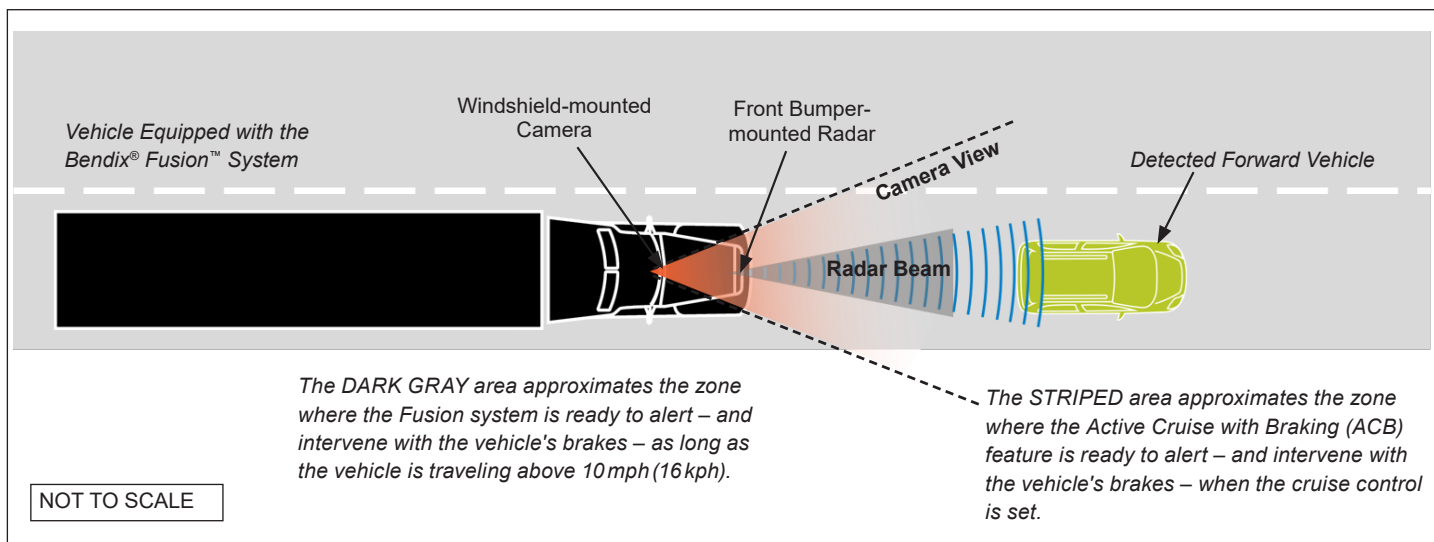


Figure 1 – Bendix® Fusion™ System

! IMPORTANT

Bendix®-brand Electronic Control Units (ECUs) are not designed to store data for purposes of accident reconstruction, and Bendix® ACom® Diagnostic Software is not intended to retrieve data for purposes of accident reconstruction. Bendix makes no representations as to the accuracy of data or video retrieved and interpreted from ECUs for purposes of accident reconstruction. Bendix does not offer accident reconstruction services or interpretation of stored data. Bendix ECUs are not protected from fire, loss of power, impact damage, or other conditions that may be sustained in a crash situation and may cause data to be unavailable or irretrievable.

! IMPORTANT

The Bendix Fusion system will not operate and will set a Diagnostic Trouble Code (DTC) if any of the following vehicle systems also show a relevant DTC: engine, engine cruise, instrument cluster, Bendix® ABS, Bendix® ATC, Bendix® ESP®, or transmission. These components must be repaired and cleared of DTCs before troubleshooting the Fusion system. NOTE: Clearing the vehicle DTCs may be the only step needed to reestablish full Fusion system functionality.

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To search DTCs and receive additional troubleshooting support, refer to:

B2Bendix.com > Services & Support > Diagnostic Trouble Code Search

To identify the camera part number, look for the Bendix® brand logo on the camera label. *Refer to Section 3.0* for how to remove the camera from the bracket to view the label.

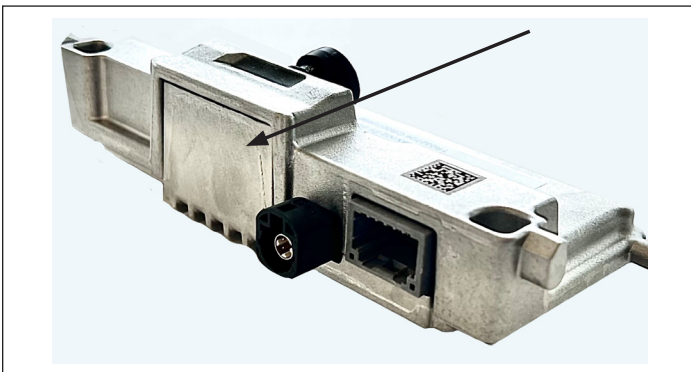


Figure 2 – Part Number Location

! IMPORTANT

For some Original Equipment Manufacturers (OEMs), the Vehicle Identification Number (VIN) of the vehicle is stored in the FLC-25's internal memory, and is cross-checked by the FLC-25 using information obtained from other vehicle controller(s). If the VIN stored in the FLC-25 does not match the VIN obtained from the other vehicle controller(s), the FLC-25 will generate a *VIN Mismatch DTC*. Refer to *B2Bendix.com > Services & Support > Diagnostic Trouble Code Search*.

GENERAL SAFETY GUIDELINES



WARNING! PLEASE READ AND FOLLOW THESE INSTRUCTIONS



TO AVOID PERSONAL INJURY OR DEATH:

When working on or around a vehicle, the following guidelines should be observed **AT ALL TIMES**:

- ▲ Park the vehicle on a level surface, apply the parking brakes, and always block the wheels. Always wear personal protection equipment.
- ▲ Stop the engine and remove the ignition key when working under or around the vehicle. When working in the engine compartment, the engine should be shut off and the ignition key should be removed. Where circumstances require that the engine be in operation, **EXTREME CAUTION** should be used to prevent personal injury resulting from contact with moving, rotating, leaking, heated, or electrically charged components.
- ▲ Do not attempt to install, remove, disassemble, or assemble a component until you have read, and thoroughly understand, the recommended procedures. Use only the proper tools and observe all precautions pertaining to use of those tools.
- ▲ If the work is being performed on the vehicle's air brake system, or any auxiliary pressurized air systems, make certain to drain the air pressure from all reservoirs before beginning ANY work on the vehicle. If the vehicle is equipped with a Bendix® AD-IS® air dryer system, a Bendix® DRM™ dryer reservoir module, a Bendix® AD-9si®, AD-HF®, or AD-HFi® air dryer, be sure to drain the purge reservoir.
- ▲ Following the vehicle manufacturer's recommended procedures, deactivate the electrical system in a manner that safely removes all electrical power from the vehicle.
- ▲ Never exceed manufacturer's recommended pressures.
- ▲ You should consult the vehicle manufacturer's operating and service manuals, and any related literature, in conjunction with the Guidelines above.
- ▲ Never connect or disconnect a hose or line containing pressure; it may whip and/or cause hazardous airborne dust and dirt particles. Wear eye protection. Slowly open connections with care, and verify that no pressure is present. Never remove a component or plug unless you are certain all system pressure has been depleted.
- ▲ Use only genuine Bendix® brand replacement parts, components, and kits. Replacement hardware, tubing, hose, fittings, wiring, etc. must be of equivalent size, type, and strength as original equipment and be designed specifically for such applications and systems.
- ▲ Components with stripped threads or damaged parts should be replaced rather than repaired. Do not attempt repairs requiring machining or welding unless specifically stated and approved by the vehicle and component manufacturer.
- ▲ Prior to returning the vehicle to service, make certain all components and systems are restored to their proper operating condition.
- ▲ For vehicles with Automatic Traction Control (ATC), the ATC function must be disabled (ATC indicator lamp should be ON) prior to performing any vehicle maintenance where one or more wheels on a drive axle are lifted off the ground and moving.
- ▲ The power **MUST** be temporarily disconnected from the radar sensor whenever any tests **USING A DYNAMOMETER** are conducted on a vehicle equipped with a Bendix® Wingman® system.

WARNING

Improper use of the Bendix® Fusion™ Driver Assistance System can result in a collision causing property damage, serious injuries, and/or death.

WARNING

Due to the inherent limitations of image recognition technology, camera-based safety technology — on rare occasions — may not be able to detect or may misinterpret lane markings. At these times, alerts may not occur, or erroneous alerts may occur.

WARNING

Bendix safety technologies complement safe driving practices. No commercial vehicle safety technology replaces a skilled, alert driver exercising safe driving techniques and proactive, comprehensive driver training. Never wait for the system to intervene. Responsibility for the safe operation of the vehicle remains with the driver at all times.

IMPORTANT

It is the responsibility of the driver to remain vigilant and change driving practices depending on traffic and road conditions.

The Bendix™ FLC-25™ camera is an integral component of the Bendix® Fusion™ Driver Assistance System, which integrates a camera, radar, and brakes. The Fusion system provides the following alerts and actions:

- Autonomous Emergency Braking (AEB)
 - Stationary Vehicle Braking (SVB)
 - Multi-lane AEB
 - Close-range AEB
- Highway Departure Braking (HDB)
- Active Cruise with Braking (ACB)
 - ACB Pedestrian Abort
- Overspeed Alert and Action (OAA)
- Alert Prioritization

NOTE: Depending on configuration, the system may record data and video of certain events and, when integrated with a telematics provider, can transmit this information for viewing on the SafetyDirect® by Rand McNally® Web Portal Processor. For support with SafetyDirect, contact Rand McNally at safetydirectsupport@randmcnally.com.

The major components that are used in the Bendix Fusion Driver Assistance System are:

- Bendix FLR-25 Radar Sensor
- Bendix® ESP® EC-80™ Controller
- SafetyDirect® Processor
- Bendix™ FLC-25™ Camera

For additional information, *refer to the following related documents on B2Bendix.com:*

- SD-13-4986, Bendix™ ESP® EC-80™ Controller
- SD-65-21025, SafetyDirect® Processor by Rand McNally®
- SD-29-50022, Bendix™ FLR-25™ Radar Sensor
- BW8107, Bendix® Safety Systems Challenging Scenarios Overview
- BW5266, Bendix® Fusion™ 3: Radar (FLR-25™) and Camera (FLC-25™) Calibration and Alignment



Figure 3 – Major Fusion System Components

1.0 INTRODUCTION TO TROUBLESHOOTING

Read the entirety of this document before performing any troubleshooting.

Diagnostic information is available using Bendix® ACom® Diagnostic Software. ACom Diagnostic Software is a PC-based program that is designed to meet RP1210 industry standards developed by the Technology and Maintenance Council (TMC). This software provides the technician with access to all available Fusion system diagnostic information and configuration capability, including:

- Electronic Control Unit (ECU) information
- Diagnostic Trouble Codes (DTCs) and repair information
- DTC reports

To search DTCs and receive additional troubleshooting support, refer to B2Bendix.com > Services & Support > Diagnostic Trouble Code Search.

For additional information on ACom Diagnostic Software and to purchase the software, refer to B2Bendix.com.

NOTE: Prior to troubleshooting, all aftermarket-installed components should be removed from the vehicle and the vehicle must be returned to factory specifications. Once the vehicle is returned to factory specifications, the fault should be recreated prior to performing troubleshooting steps.

IMPORTANT

All vehicle DTCs related to the engine, transmission, instrument cluster, engine cruise control, and Bendix® Antilock Braking System (ABS), Automatic Traction Control (ATC), or Electronic Stability Program (ESP®) systems must first be resolved with no DTCs present during the vehicle operation while in cruise control before trying to resolve camera DTCs.

CAUTION

If a problem with the FLC-25 camera is detected, reduced performance may occur, up to and including loss of Automated Emergency Braking (AEB). The vehicle should be serviced as soon as possible to restore full functionality.

1.1 TEMPORARILY DISABLE THE LANE DEPARTURE WARNING (LDW)

To avoid erroneous LDW warnings in areas such as construction zones – where the non-standard or overlapping road markings present might cause false alerts – the Fusion Active Safety System has an enable/disable switch. The Bendix switch design or a similar OEM switch may be used. See Figure 4.



Figure 4 – Example LDW Enable/Disable Switch

For vehicles that have the LDW switch hard-wired to the SafetyDirect® by Rand McNally® Web Portal Processor, and a functioning On Board Computer (OBC)/Telematics system: The enable/disable switch used by the LDW system also functions – when depressed for six (6) seconds – to activate a manual request to transmit the last five (5) seconds and next five seconds of buffered video data.

1.2 SETTING DIAGNOSTIC TROUBLE CODES (DTCs)

If, during operation, the Fusion system detects a problem with the FLC-25 camera, a DTC will be set and – depending on the OEM – the driver will be alerted on the dash display with an icon or similar method. In these cases, some features of the Fusion system will not be available.

For example, if – for an extended period of time – the system detects that the camera is blocked by dirt, snow, ice, etc., a DTC will typically be set.

2.0 TROUBLESHOOTING AND DIAGNOSTICS



All vehicle DTCs related to the engine, transmission, instrument cluster, engine cruise control, and Bendix ABS, ATC, or ESP systems must first be resolved—with no DTCs present during the vehicle operation while in cruise control—before attempting to diagnose Fusion system DTCs.

Important examples are VDC2 and EBC DTCs which are typically related to the brake controller. The hierarchy of ECUs determines that any DTCs on the brake controller must be resolved before attempting to troubleshoot the Fusion system.

2.1 TROUBLESHOOTING DTCs: POWER SUPPLY

IGNITION VOLTAGE TOO LOW

1. Measure the ignition voltage under load.
2. Ensure that the ignition voltage is greater than 9 VDC.
3. Check the vehicle battery and associated components. Inspect for damaged wiring, damaged or corroded connectors, and loose connections.
4. Check the condition of the fuse.

IGNITION VOLTAGE TOO HIGH

1. Measure the ignition voltage.
2. Ensure that ignition voltage is not greater than 16 VDC.
3. Check the vehicle battery and associated components.
4. Inspect for damaged wiring, damaged or corroded connectors, and loose connections.

CONNECTOR PIN-OUT AND POWER REQUIREMENTS

The camera has two (2) connectors. See Figure 5.

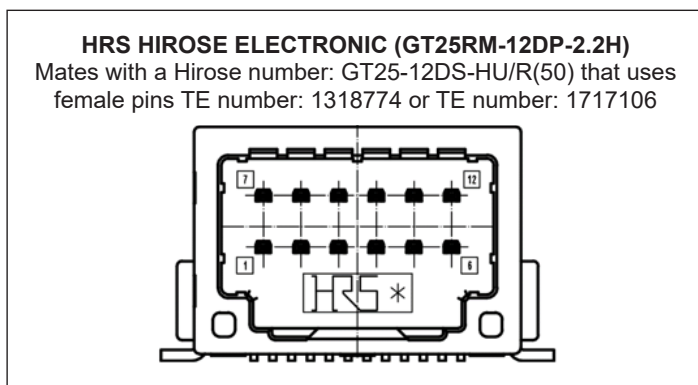


Figure 5 – Camera and Harness Connector Views

Pin #	Description	Nominal Voltage	Nominal Current
1	Not Used		
2	Private Communications Low	5 V	
3	J1939 Low	5 V	
4	Ground GND (-) Optional	0 V	540 mA
5	Not Used		
6	Not Used		
7	Module Power	14 V	540 mA
8	Private Communications High	5 V	
9	J1939 High	5 V	
10	Ground GND (-)	0 V	540 mA
11	Not Used		
12	Not Used		

Table 1 – Harness Connector Pins

POWER SUPPLY TESTS

1. Take all measurements at the camera's harness connector.
2. Place a load (e.g. 1157 stop lamp) across the supply voltage and ground connection. Measure the voltage with the load. The supply voltage on pin 7 to ground should measure between 10 to 16 VDC.
3. Check for damaged wiring, damaged or corroded connectors, and loose connections.
4. Check the condition of the vehicle battery and associated components. Ensure the connection to ground is secure and tight.
5. Using the procedures described by the vehicle manufacturer, check the alternator output for excessive noise.

2.2 COMMUNICATIONS (J1939 AND PRIVATE) TEST AND TROUBLESHOOTING PROCEDURES

The Bendix FLC-25 camera requires private network messages to and from the Fusion system. The camera will set a DTC if these messages are not present, or if there is a problem with the private communications system. *Refer to SD-29-50022, Bendix® Fusion™ FLR-25™ Radar Sensor, on B2Bendix.com* for full troubleshooting information.

NOTE: Confirm there are no DTCs on other components. If there are active DTCs, they should be addressed prior to performing additional troubleshooting procedures.

1. Take all measurements at the harness connector unless otherwise indicated.



Do not insert any probe into the pin on the mating connector of the camera that is greater than the width of a terminal. Damaged connector pins will require the replacement of the harness.

2. Check for damaged or reversed private communications wiring.

If the private communications HIGH **or** private communications LOW wiring circuits are damaged, such as shorting together, the entire private communications link will be lost. The problem may be intermittent, enabling the private communications link to operate normally sometimes. In this event, multiple DTCs may be logged in the camera and radar.

If the private communications HIGH **and** private communications LOW wiring circuits are reversed, communication over the entire private communications link will be lost. Devices that use the affected network will not be able to transmit or receive messages on that network.

3. Check for corroded or damaged wiring connector problems such as opens or shorts to voltage or ground.
4. If the connector terminals are corroded, this may be an indication of water intrusion into the wiring system and possibly the camera. Replacement of the entire harness is recommended. If the terminals of the camera are corroded, replacement of the camera is recommended.

5. Check for other private communications devices which may be inhibiting communication. The service technician should consult the procedures for private communications troubleshooting. The device's power should be removed and measurements made at the ECU pins for shorts to ground and power pins and resistance between the private communications HIGH or private communications LOW input circuits.
6. Unplug the camera harness. With the ignition switch off, measure the resistance (ohms) using a multimeter.
 - Private Communications: Measure the resistance between harness pins 2 and 8. The reading should be approximately 120 ohms. If it is not, the vehicle wiring should be investigated.
 - J1939: Measure the resistance between harness pins 3 and 9. The reading should be 0 ohms. If it is not, the vehicle wiring should be investigated.

2.3 PRIVATE COMMUNICATIONS NETWORK TEST PROCEDURE

The Bendix FLC-25 camera requires private network messages to and from the Fusion system. The camera will set a DTC if these messages are not present, or if there is a problem with the private communications system. *Refer to SD-29-50022, Bendix® Fusion™ FLR-25™ Radar Sensor, on B2Bendix.com* for full troubleshooting information.

2.4 TROUBLESHOOTING WIRING HARNESSES

All wire harness connectors must be properly seated to maintain electrical connectivity. Push the mating connectors until they click. When replacing an FLC-25 camera, check that the wire harness connectors are free of damage, including corrosion, before plugging into a new camera. Check for corroded or damaged wiring connector problems such as opens or shorts to voltage or ground.

If the connector terminals are corroded, this may be an indication of water intrusion into the wiring system and possibly into the camera (presumably from a cracked windshield). Replacement of the entire harness is recommended. If the terminals of the camera are corroded, replacement of the camera is recommended.

3.0 INSTALLATION AND REPLACEMENT

! IMPORTANT

! WARNING

Whenever reinstalling or replacing a camera – for example, after a windshield is replaced – the OEM-recommended position for the vehicle must be used. Failure to install the camera in the correct position can result in system DTCs being set, system performance degradation, and/or a potential collision causing property damage, serious injuries, and/or death.

The Bendix FLC-25 camera is installed on the windshield at a position determined by the OEM with support from Bendix engineering. When reinstalling or replacing a camera on an OEM windshield, follow all OEM guidelines. Below is the Bendix understanding of mounting positions for vehicles without an OEM windshield. Always refer to OEM documentation for most accurate and up-to-date information. See Figure 6 and refer to Table 2 for mounting guidelines for cameras on a non-OEM windshield.

NOTE: When replacing a camera bracket, temporarily mark the location of the top of the original bracket to help position the replacement.

Use only OEM installation methods or Bendix-approved windshield adhesive when remounting a camera bracket (included with the bracket kit).

The ambient temperature must be in the range of 50-100° F (10-38° C). Thoroughly clean the area of the windshield where the camera will be installed with a lint-free cloth and a 50/50 water/isopropyl alcohol solution.

Make certain that there is no grease or contamination present and that the windshield is completely dry before installing the bracket.

Use removable tape or a non-permanent marker to indicate where the top of the bracket will be installed. Remove the protective film covering the tape adhesive on the bracket and, using a small “torpedo” level to be sure that it is level, install the bracket on the glass, holding firmly. Use a seam roller (or similar tool) to apply pressure to the back side of the bracket opposite the tape. If using a seam roller, pass it firmly over the taped area two (2) or three (3) times. Wait at least 20 minutes before installing the camera, at which point a 50% bond strength is created. The full bond between the bracket and windshield is achieved after 72 hours.

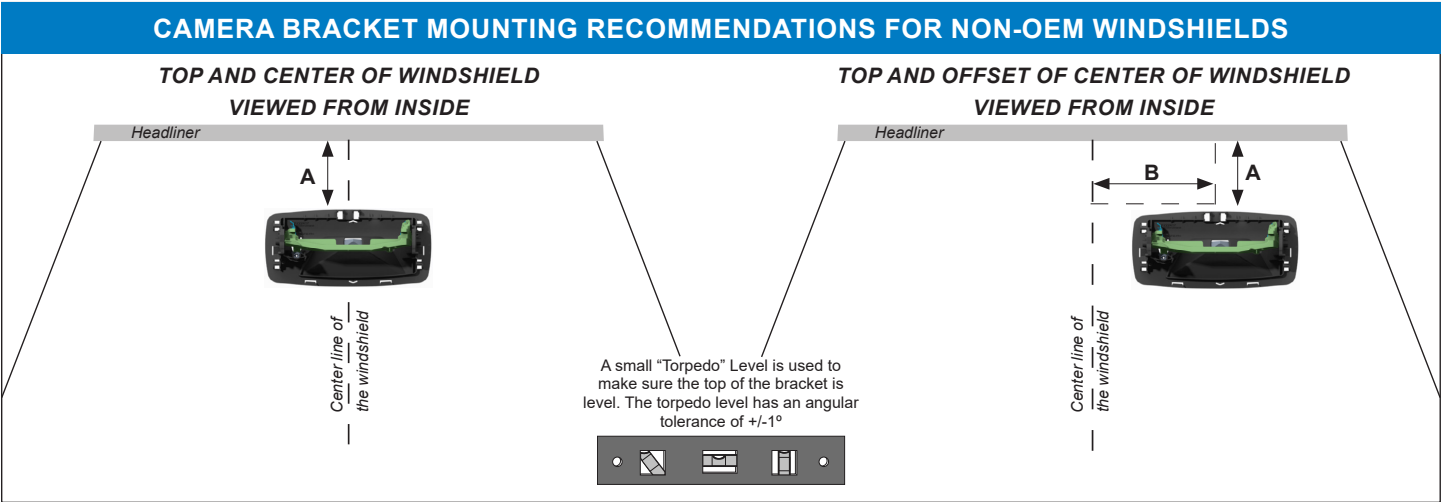


Figure 6 – Camera Bracket Mounting Recommendations for NON-OEM Windshields

CAMERA BRACKET MOUNTING RECOMMENDATIONS FOR NON-OEM WINDSHIELDS				
Vehicle OEM*	Vehicle Model	Distance “A”	Distance “B”	COMMENTS
Kenworth®	T680	0.75 in. (19mm)	0	Distance A: measured from headliner All distances are ± ⅛ in. (3 mm)
International®	HX® Series, LT® Series, RH™ Series, MV	2.5 in. (63.5 mm)	0	
			0	
International® IC Bus™	CE Series	7.125 in. (180 mm)	6 in. (152mm)	
Peterbilt®	579	0.75 in. (19mm)	0	
Battle Motors	Striker	5.2 in. (133mm)	7.87 in. (200mm)	Distance A: measured from rubber seal
	Raider	7.4 in. (188mm)		All distances are ± ⅛ in. (3 mm)

* All trademarks shown here are the property of their respective owners and are used for reference only.

Table 2 – Camera Mounting from Top of Windshield Specifications

COVER AND CAMERA REMOVAL

The camera is most easily removed from the holder if it is removed from the bracket assembly. This will depend upon the proximity of the headliner. In most cases, there will not be sufficient clearance to access the clips that retain the camera in the holder.

1. Begin by removing the cover from the camera. This is the reverse of installation. Push the cover up. Once the cover is moved upward, it can be pulled back, and off the bracket assembly. Take note of the holder position.
2. Remove the electrical connectors with attached wiring from the camera. The connectors have locking features that require squeezing them to facilitate removal.
3. See *Figure 7*. Use a screwdriver to pry the locking beam (A) away from the hub (B) on the right side of the bracket.

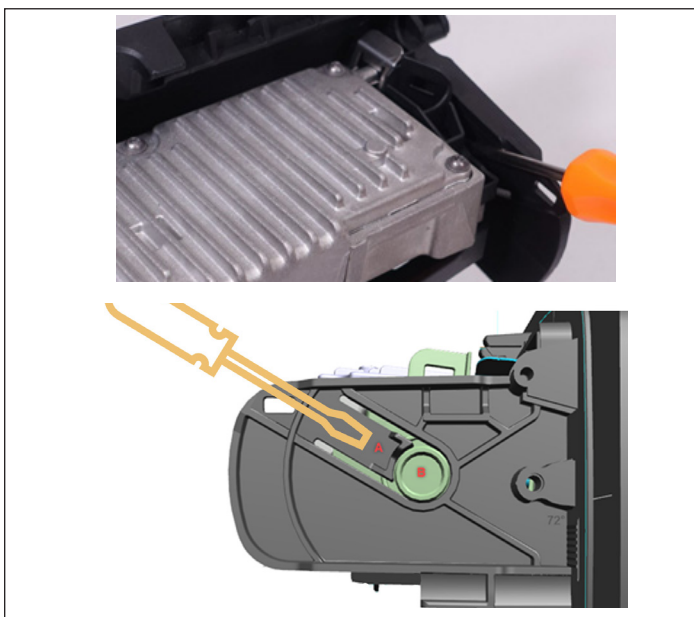


Figure 7 – Camera Removal

4. Once the hub (B) is released from the beam (A), the right side of the holder with the camera can be pulled back from the bracket assembly. This will allow the entire holder and camera to be removed from the bracket assembly. See *Figure 8*. **NOTE:** Only the holder and bracket assembly are shown.

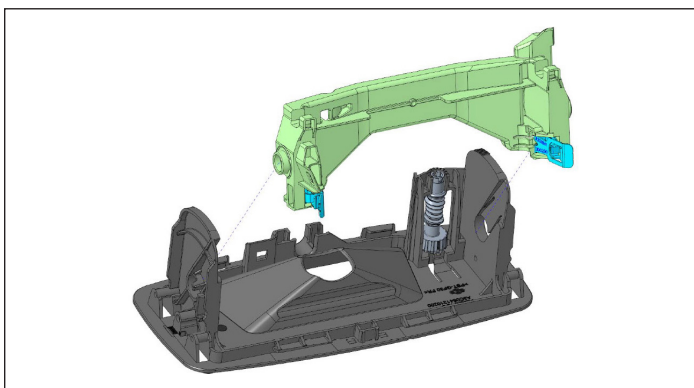


Figure 8 – Holder Removal

5. Once on the bench, the clips can be released from the tabs on the camera body. Push on the clips as indicated by the arrows in *Figure 9*.



Figure 9 – Camera Removal

6. Once the clips are released, the camera can be rotated up and out of the clips. This may require the use of screwdrivers to pry the camera from the clips. See *Figure 10*.

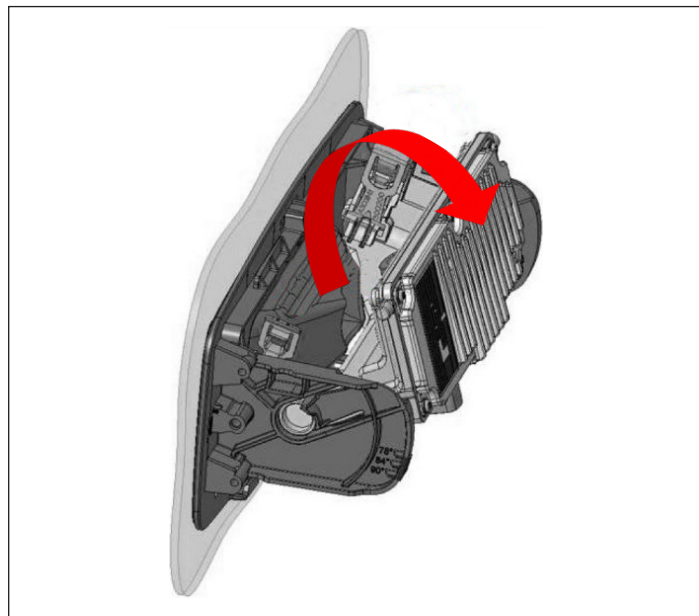


Figure 10 – Camera Rotation

WARNING

Do NOT pry between the bottom of the camera and the holder with tools. It is easily broken.

7. Check the holder for damage. If there is no damage, return the holder to the bracket. Note that the hubs on the holder fit into slots on the bracket and then can be slid into place. Start with the hub on the right (curb) side of the bracket. Once it is seated, push the other side home, engaging the teeth on the holder with the threads on the worm shaft.

8. It may be necessary to adjust the holder position back to its initial position as previously noted (*Step 1*).

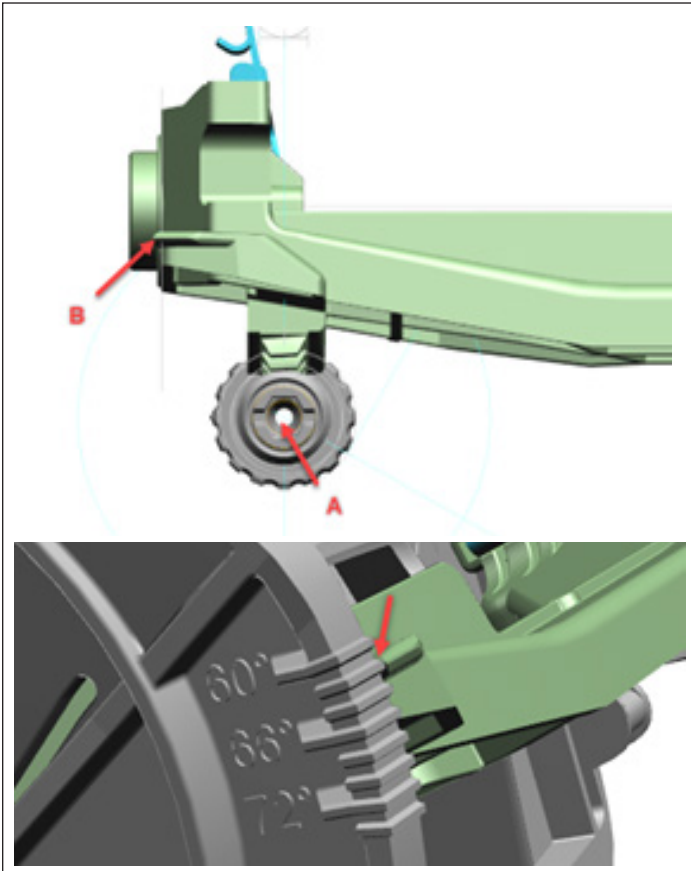


Figure 11 – Worm Shaft Adjustment

CAMERA INSTALLATION

In instances where a third-party bracket and/or cover is used, follow the applicable OEM instructions. If a Bendix bracket and/or cover is used, refer to the following procedures.

NOTE: A Bendix bracket and cover can be identified by a Knorr-Bremse® logo on the component. See *Figure 12*.

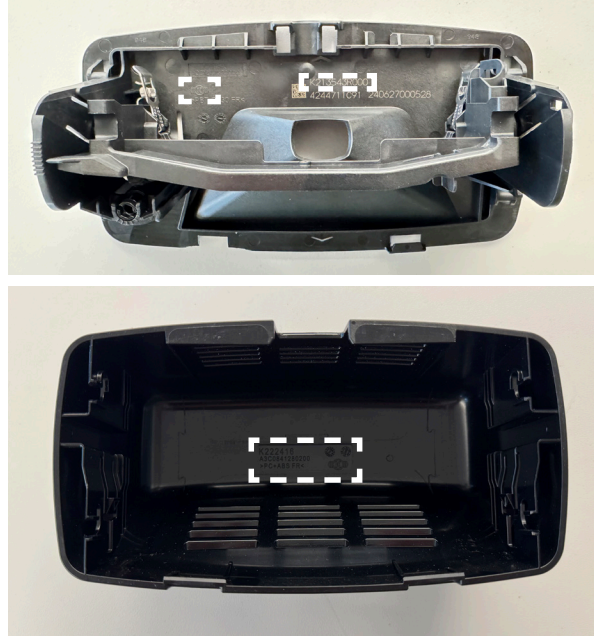


Figure 12 – Bendix Bracket and Cover PN Location

1. Hold the camera in a lens-up orientation. Referring to *Figure 13*, insert the tabs (A) into the slots (B).

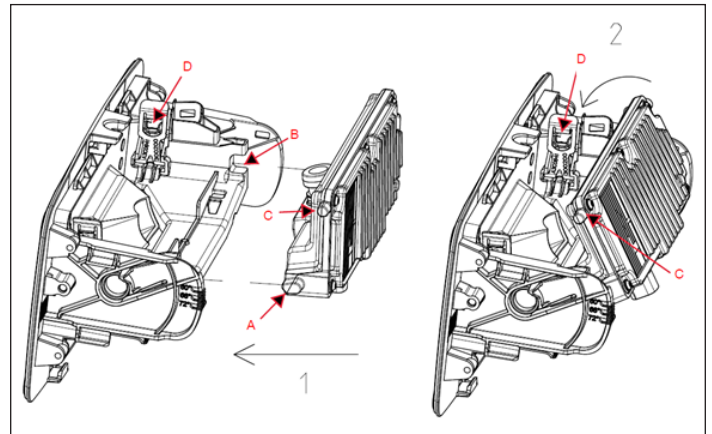


Figure 13 – Camera Installation

Rotate the camera counterclockwise (when viewed from the left [driver] side) until the tabs (C) contact the clips (D).

-
- A close-up photograph of a silver metal electronic component, possibly a sensor or actuator, mounted on a black plastic housing. The component features a QR code and text including "industrial", "Made in Germany", and "D-41507 Appeler". The component is secured with two screws.

3. In some cases, the clips will not “click” into position. Verify that the clips are in the locked position as shown in *Figure 15*. Sometimes it is necessary to pry the clips into the final locked position. In most cases, this can be done using finger pressure. In some cases, needle-nose pliers or small screw drivers may be needed.



NOTE: Due to proximity to the headliner, it may be necessary to use an inspection mirror to verify the clips are locked.

The bracket should be removed and installed per OEM recommendations.

See Figure 16.

-

Figure 16 – Cover Installation

3.1 CAMERA CALIBRATION

Calibration is done as part of initial drive for service cameras. It may take up to 20 minutes of driving above 25 mph (40 kph) on well-marked roads in clear weather for the camera to be ready and fully calibrated. The camera should not be returned for any calibration faults found in this time period. If a calibration fault appears while driving, check the camera for looseness or misalignment and continue driving to clear the fault.

3.2 DASH SWITCH/LAMPS

Each OEM has their own method for displaying the system status to the driver, and there will typically will be a switch on the dash to allow the driver to temporarily disable the system. *Refer to the OEM Operator's Manual* for system indicator lamp(s) used for the FLC-25 camera, and whether there is a bulb-check illumination at vehicle power-up.

Additionally, the system self-monitors and will set a DTC that will typically alert the driver using a similar icon on the dash display or by a status lamp. *Refer to Section 1.0: Introduction to Troubleshooting* for more information.

3.3 MAINTENANCE

In normal use, the FLC-25 camera needs only a clean, properly maintained windshield to ensure a clear view of the road ahead. Protect the camera lens whenever the inside of the windshield is cleaned.

3.4 CAMERA INTERCHANGEABILITY

When replacing cameras only, use replacements with the same part number (or a direct superseding replacement number supplied by Bendix). After replacing the camera, run the engine for two (2) minutes and verify no faults are present. Once confirmed no faults are shown with the engine running, drive the vehicle for 5-20 miles (8-32 km) or until the camera calibration fault is no longer indicated.



Cameras of different vehicle models and model years must not be interchanged. The use of an incorrect camera can lead to DTCs being set, and performance degradation – including unnecessary system interventions and the potential for situations where interventions do not occur when they would normally.

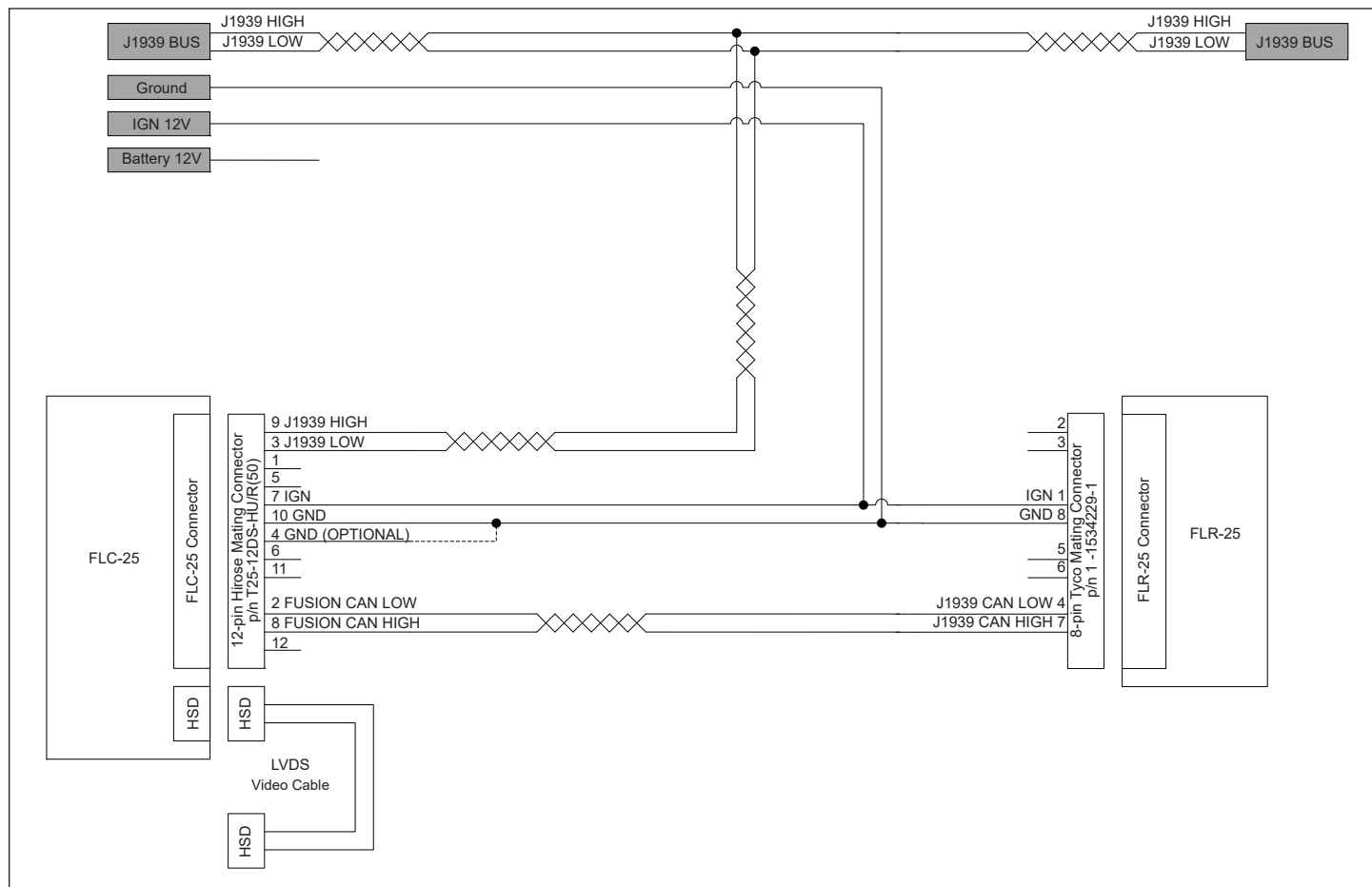
Cameras are designed specifically for a particular vehicle and model. DTCs caused by relocating cameras to an incorrect vehicle may result in the vehicle system using the camera to be partially or fully unavailable.

TECHNICAL ASSISTANCE

For additional assistance, the Bendix Tech Team can be reached by email at techteam@bendix.com or by phone at 1-800-AIR-BRAKE (1-800-247-2725), option 2. Bendix Tech Team representatives are available Monday through Thursday, 8:00 a.m. – 6:00 p.m., and Friday, 8:00 a.m. – 5:00 p.m. ET.

APPENDIX A – SYSTEM WIRING SCHEMATIC

BENDIX FUSION 3.0: FLR-25 AND FLC-25 WIRING SCHEMATIC



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